

13.8 LAB: Digital forensics (Walkthrough)

(Page 1 of 4)

Introduction

Digital forensics is a branch of forensic science focused on recovering, collecting, analyzing, and examining data found in devices. Different forensic techniques include recovering deleting files, analyzing browser activity, and examining metadata.

Conducting forensic analysis first requires a complete copy of a device. The copy must be bit-for-bit to ensure accuracy. The Linux command **dd** is used to make a bit-for-bit copy.

FTK Imager (Forensics Toolkit) is a Windows tool that copies a system similar to **dd**. The toolkit can also analyze emails, decrypt files, and detect malware. **E01** is a forensic image file for a software application called EnCase. Compared to **dd** images, **E01** takes up less disk space and contains metadata.

Learning Objectives

- Create a forensics copy of a system
- View a forensics copy of a system using Autopsy
- Recover and analyze deleted files
- Generate a forensics report

Devices

The lab contains one virtual machine:

- **ZYWIN01** (Windows Server 2022)

Tasks

Task 1: Convert a **dd** image to **E01** using FTK Imager

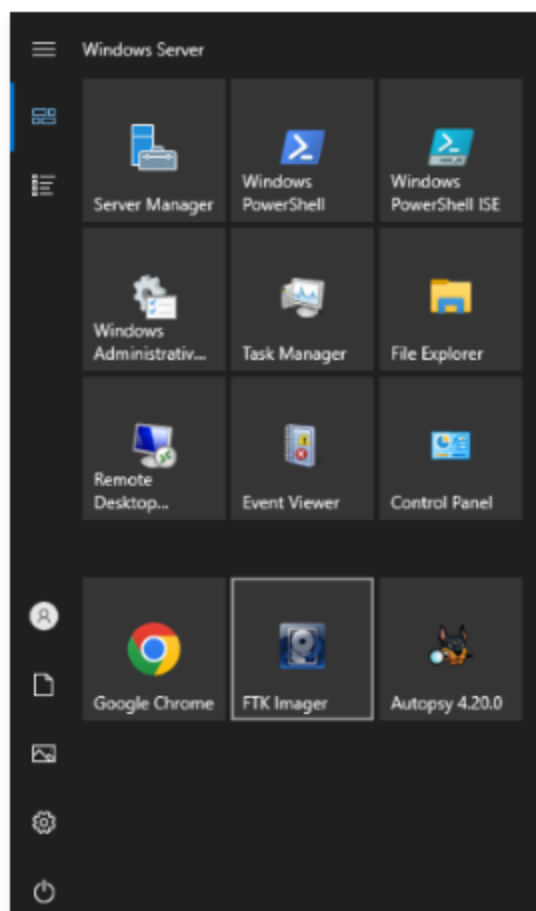
Task 2: Use Autopsy to analyze the file system

Task 3: Generate a report with Autopsy

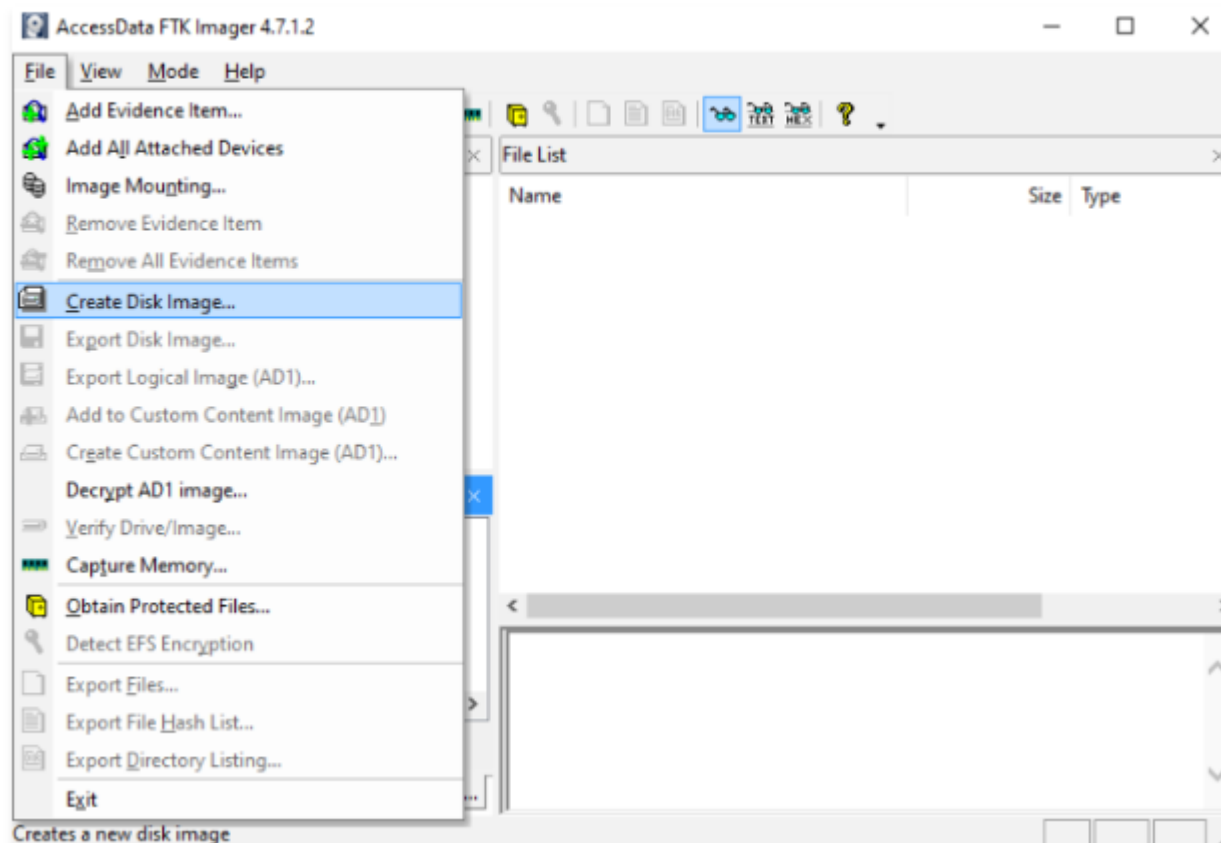
Task 1: Convert a dd image to E01 using FTK Imager

1. Select **ZYWIN01** from the server drop-down menu in the right pane and click **Connect to VM**.

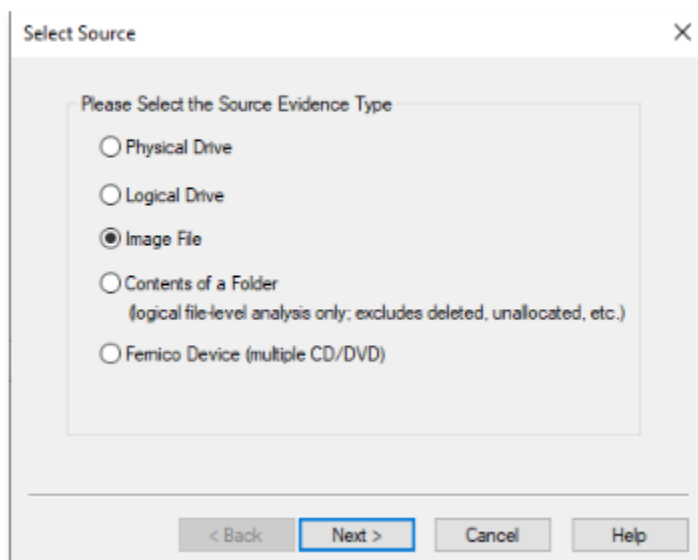
2. Select the **Windows Start Menu** at the bottom left. Select **FTK Imager**.



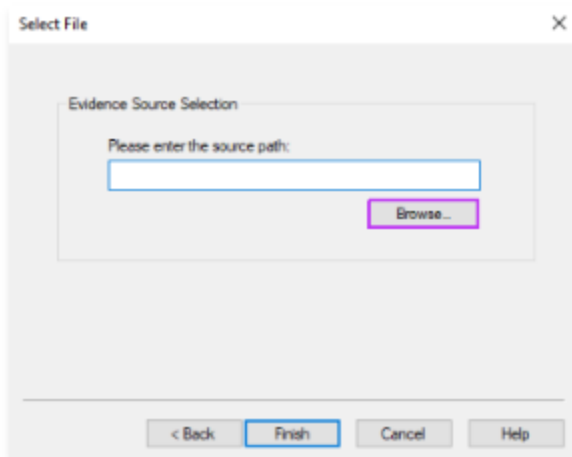
3. Select **File** → **Create Disk Image**.



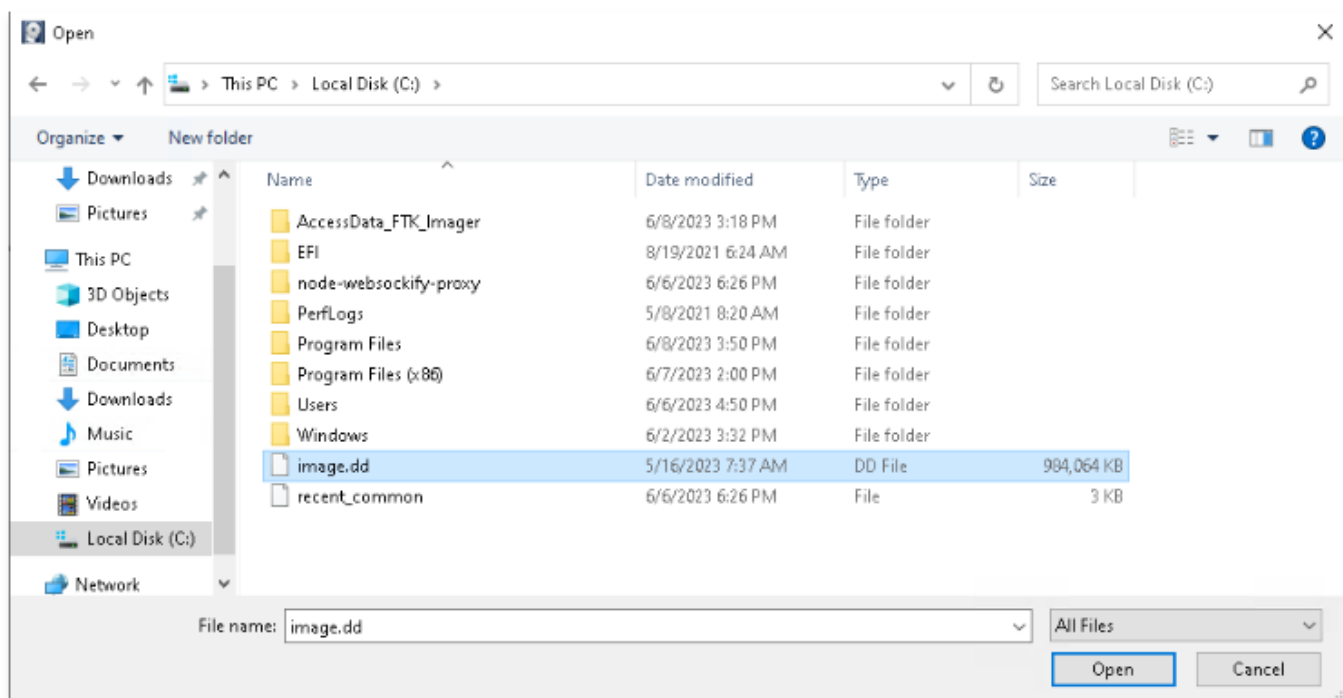
4. Select **Image File** and then **Next**.



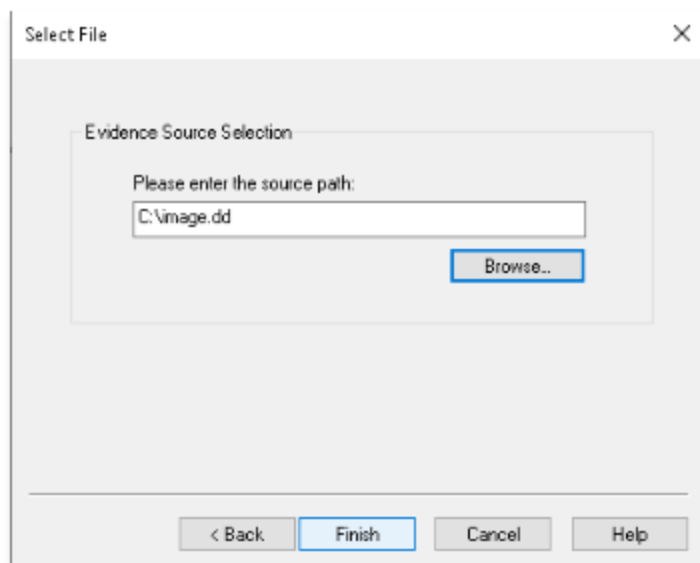
5. Click **Browse**.



6. Go to **This PC → Local Disk (C:)**. Select **image.dd** and click **Open**.

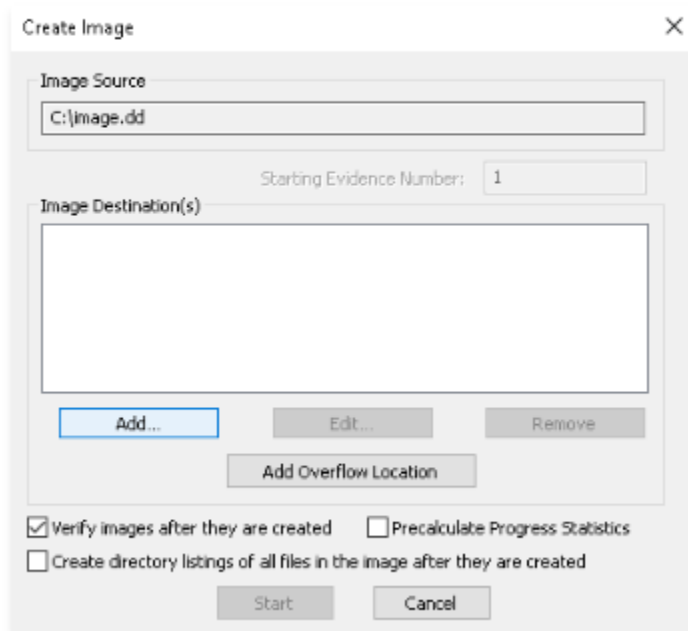


7. Click **Finish**.



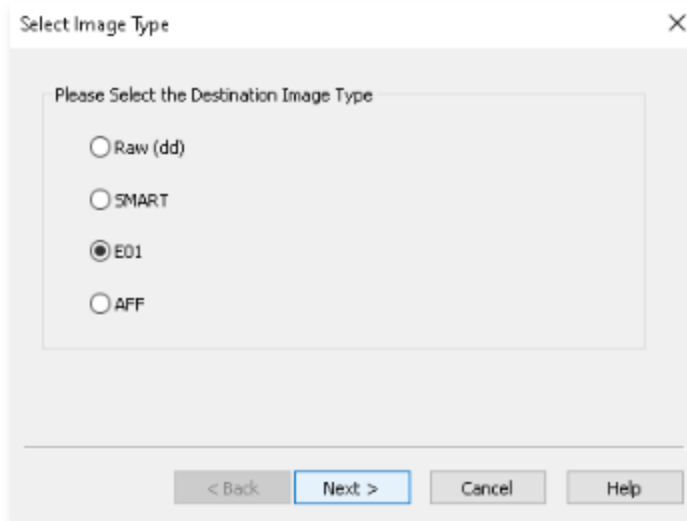
The 'Select File' dialog box is shown with a close button (X) in the top right corner. It contains a section titled 'Evidence Source Selection' with the instruction 'Please enter the source path:'. Below this is a text input field containing 'C:\image.dd' and a 'Browse...' button. At the bottom of the dialog are four buttons: '< Back', 'Finish', 'Cancel', and 'Help'.

8. On **Create Image**, Click **Add**.



The 'Create Image' dialog box is shown with a close button (X) in the top right corner. It contains an 'Image Source' section with a text input field containing 'C:\image.dd'. Below this is a 'Starting Evidence Number' field with the value '1'. The 'Image Destination(s)' section contains a large empty text area. Below this area are three buttons: 'Add...', 'Edit...', and 'Remove'. Below these is an 'Add Overflow Location' button. At the bottom are three checkboxes: 'Verify images after they are created' (checked), 'Precalculate Progress Statistics' (unchecked), and 'Create directory listings of all files in the image after they are created' (unchecked). At the very bottom are 'Start' and 'Cancel' buttons.

9. Choose **E01** to select the destination image type. Click **Next**.

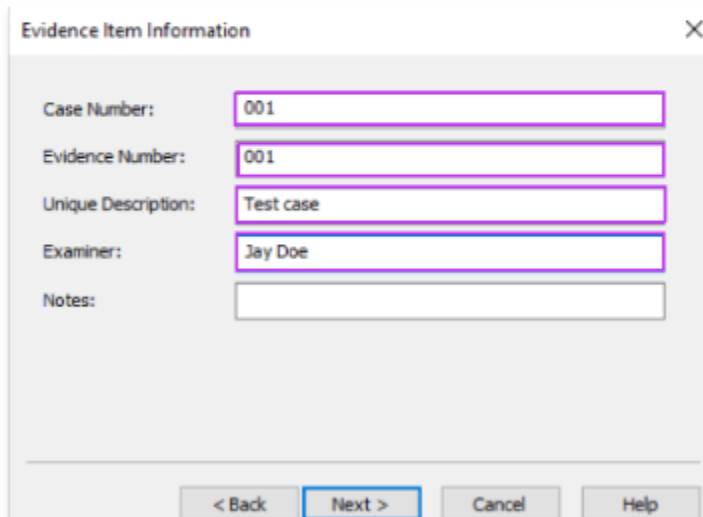


A dialog box titled "Select Image Type" with a close button (X) in the top right corner. Inside the dialog, there is a section titled "Please Select the Destination Image Type" containing four radio button options: "Raw (dd)", "SMART", "E01", and "AFF". The "E01" option is selected, indicated by a filled circle. At the bottom of the dialog, there are four buttons: "< Back", "Next >", "Cancel", and "Help". The "Next >" button is highlighted with a blue border.

10. Enter the following information:

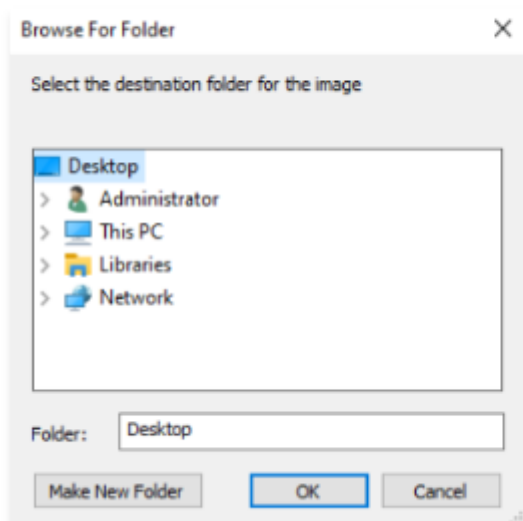
- **Case Number:** 001
- **Evidence Number:** 001
- **Unique Description:** Test case
- **Examiner:** Jay Doe

Click **Next**.

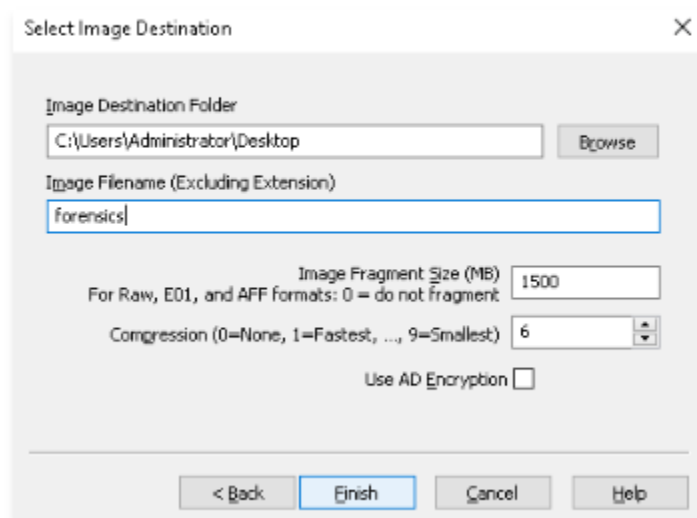


A dialog box titled "Evidence Item Information" with a close button (X) in the top right corner. Inside the dialog, there are five input fields, each with a label to its left: "Case Number:" with value "001", "Evidence Number:" with value "001", "Unique Description:" with value "Test case", "Examiner:" with value "Jay Doe", and "Notes:" with an empty field. At the bottom of the dialog, there are four buttons: "< Back", "Next >", "Cancel", and "Help". The "Next >" button is highlighted with a blue border.

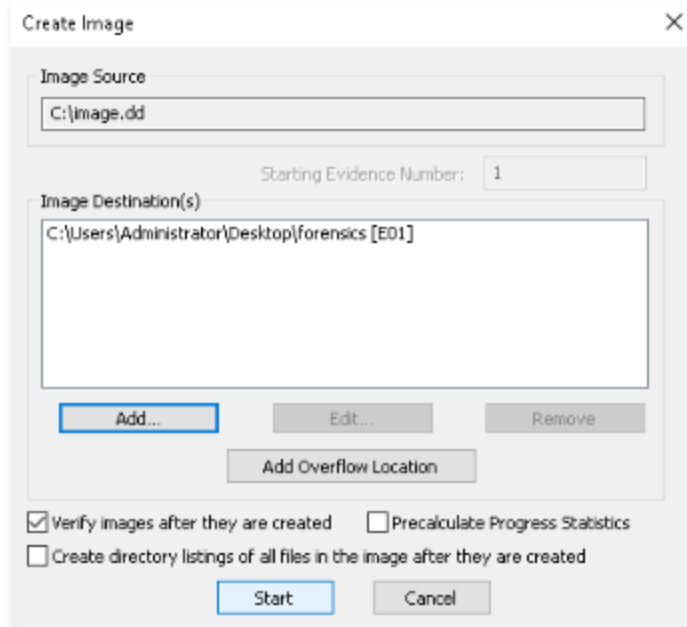
11. For image destination, click **Browse**. Select **Desktop** and click **OK**.



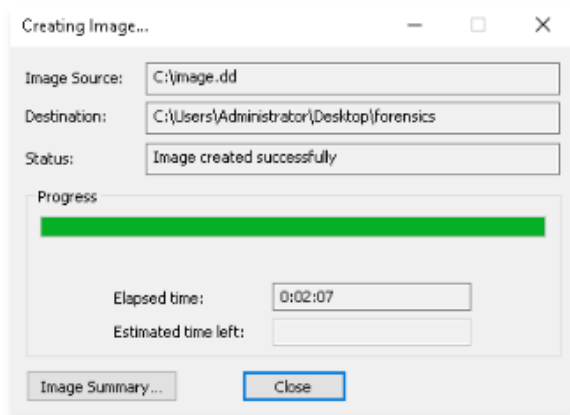
12. Enter **forensics** for **Image Filename** and click **Finish**.



13. Click **Start**.



14. Once the image is created, click **Close** on the **Drive/Image Verify Results** and **Creating Image...** windows. (Note: Image creation may take a few minutes to complete.)



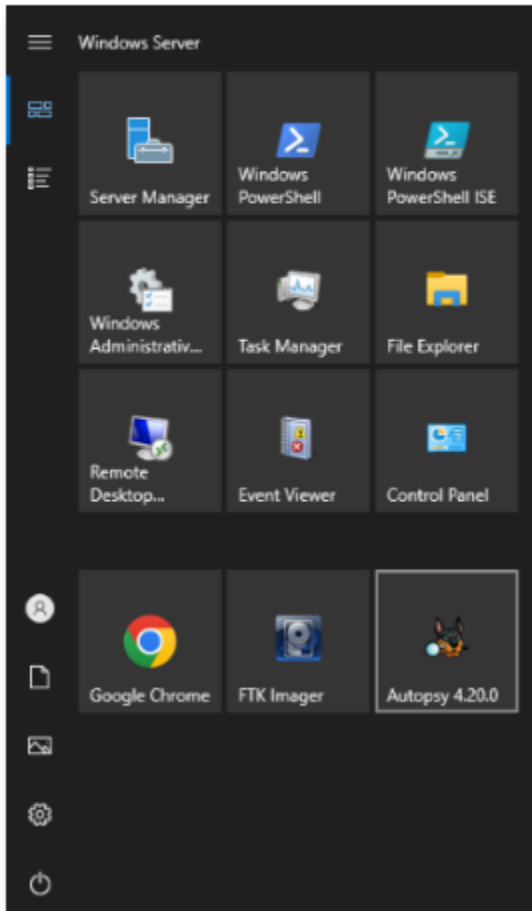
Points

0/3  E01 image created.

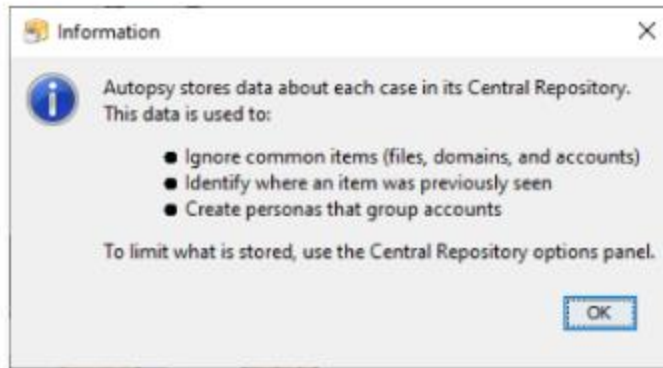
15. **Close** FTK Imager.

Task 2: Use Autopsy to analyze the file system

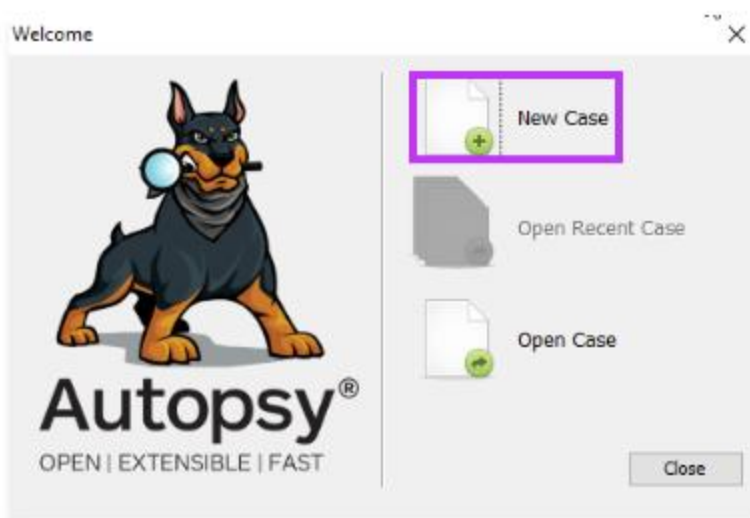
1. Open the **Windows Start Menu**. Click **Autopsy 4.20.0**.



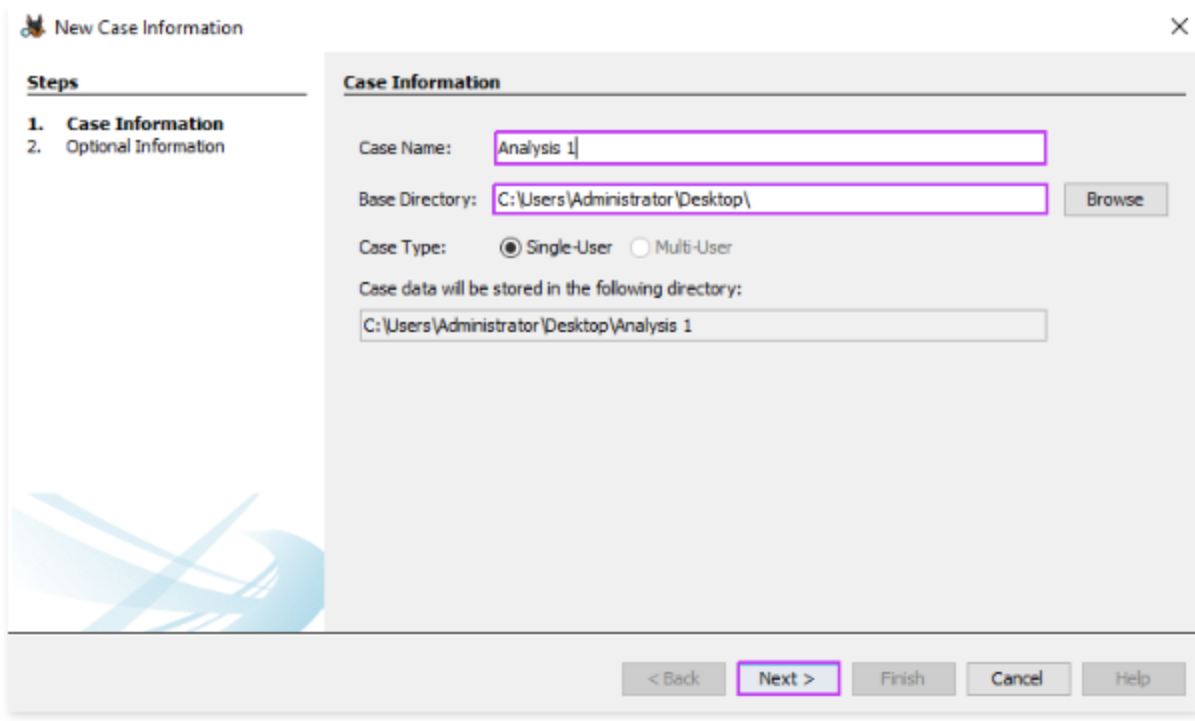
2. Click **OK**.



3. On the welcome screen, select **New Case**.



4. Enter **Analysis 1** for **Case Name**. For Base Directory, browse to select **Desktop**. Click **Next**.

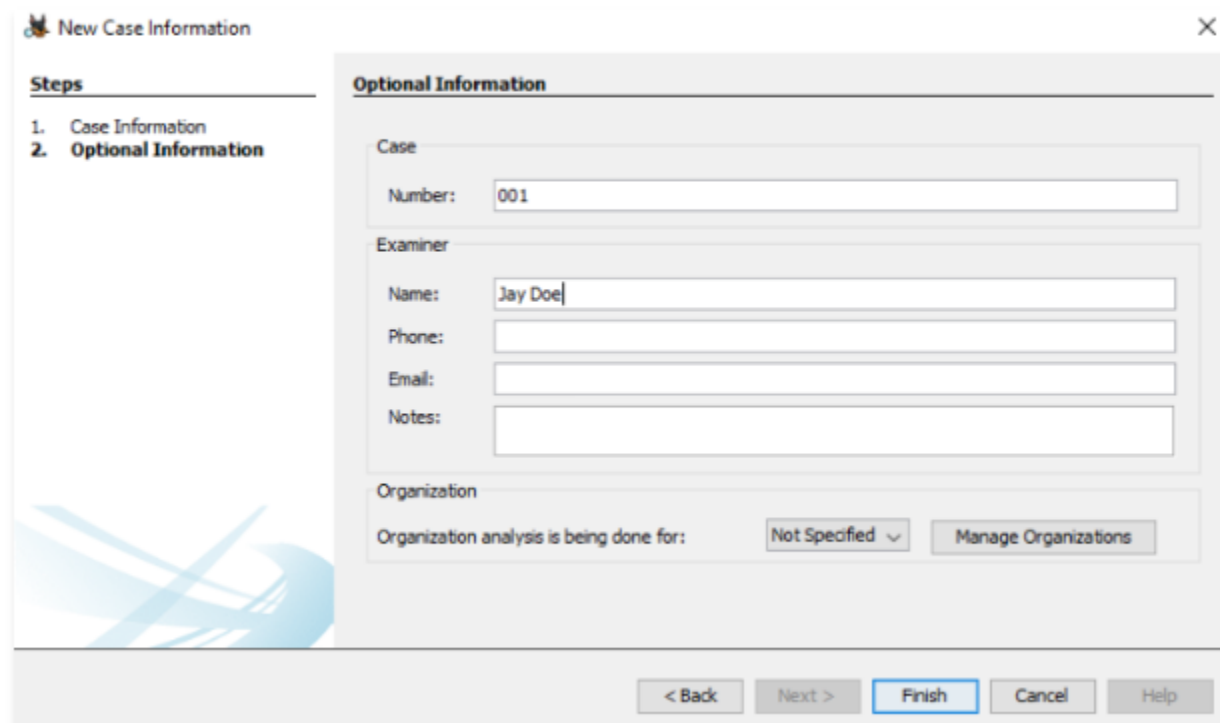


The dialog box is titled "New Case Information" and has a close button (X) in the top right corner. On the left, a "Steps" pane shows two steps: "1. Case Information" (selected) and "2. Optional Information". The main area is titled "Case Information" and contains the following fields and controls:

- Case Name:** A text box containing "Analysis 1".
- Base Directory:** A text box containing "C:\Users\Administrator\Desktop\" with a "Browse" button to its right.
- Case Type:** Two radio buttons: "Single-User" (selected) and "Multi-User".
- Case data will be stored in the following directory:** A text box containing "C:\Users\Administrator\Desktop\Analysis 1".

At the bottom, there are five buttons: "< Back", "Next >" (highlighted with a red border), "Finish", "Cancel", and "Help".

5. For **Number**, enter **001** . For **Name**, enter **Jay Doe**. Click **Finish**.



The dialog box is titled "New Case Information" and has a close button (X) in the top right corner. On the left, a "Steps" pane shows two steps: "1. Case Information" and "2. Optional Information" (selected). The main area is titled "Optional Information" and contains the following fields and controls:

- Case:** A section containing a "Number:" text box with "001".
- Examiner:** A section containing "Name:" (text box with "Jay Doe"), "Phone:" (text box), "Email:" (text box), and "Notes:" (text box).
- Organization:** A section containing "Organization analysis is being done for:" with a "Not Specified" dropdown menu and a "Manage Organizations" button.

At the bottom, there are five buttons: "< Back", "Next >", "Finish" (highlighted with a blue border), "Cancel", and "Help".

6. Select **Generate new host name based on data source name**. Click **Next**.

The screenshot shows a wizard window titled "Add Data Source" with a close button (X) in the top right corner. On the left, a "Steps" sidebar lists five steps: 1. Select Host (highlighted), 2. Select Data Source Type, 3. Select Data Source, 4. Configure Ingest, and 5. Add Data Source. The main area is titled "Select Host" and contains the text "Hosts are used to organize data sources and other data." Below this text are three radio button options: "Generate new host name based on data source name" (which is selected and has a dashed border), "Specify new host name" (with an adjacent text input field), and "Use existing host" (with an adjacent large empty rectangular box). At the bottom right, there are five buttons: "< Back", "Next >" (highlighted with a blue border), "Finish", "Cancel", and "Help".

Add Data Source

Steps

- Select Host**
- Select Data Source Type
- Select Data Source
- Configure Ingest
- Add Data Source

Select Host

Hosts are used to organize data sources and other data.

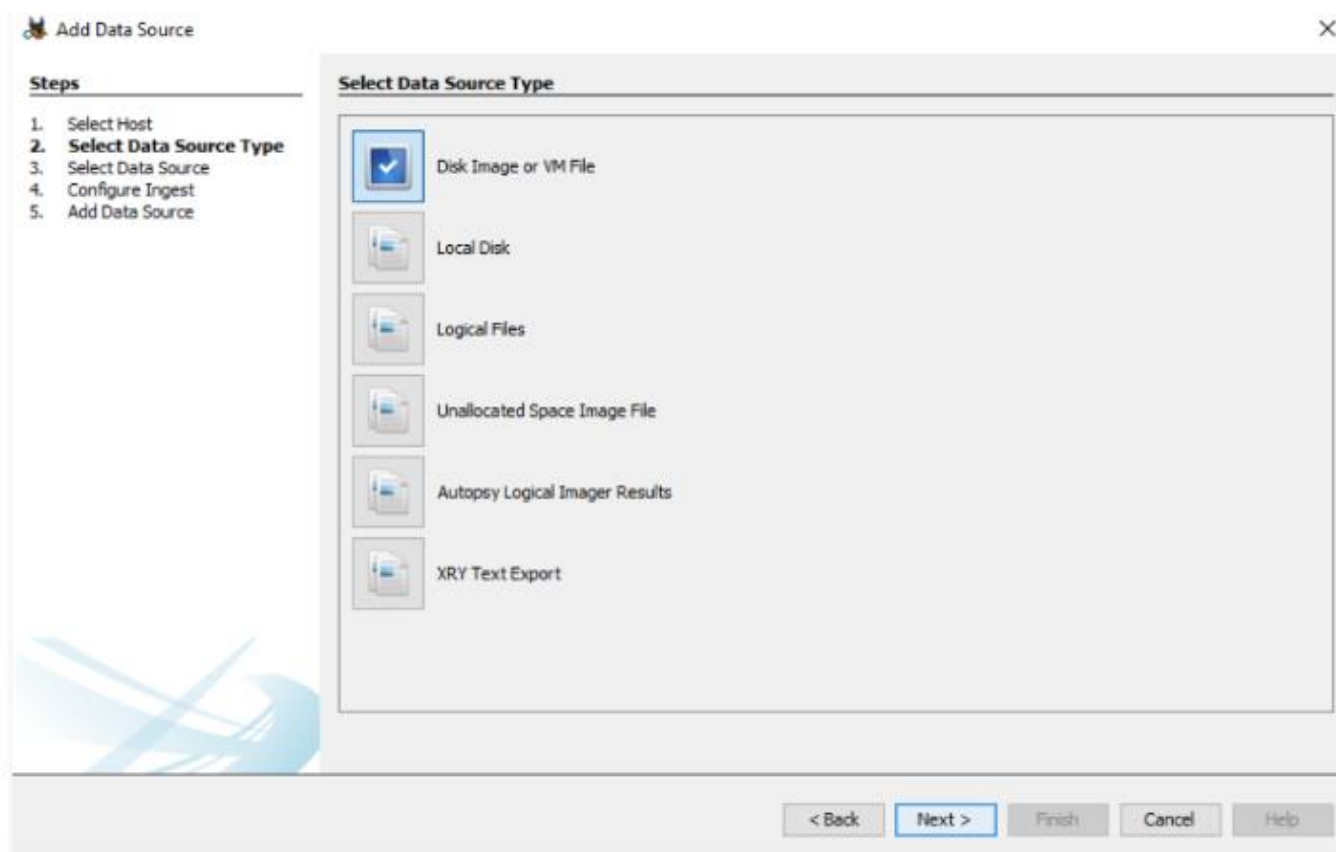
☒ Generate new host name based on data source name

☐ Specify new host name

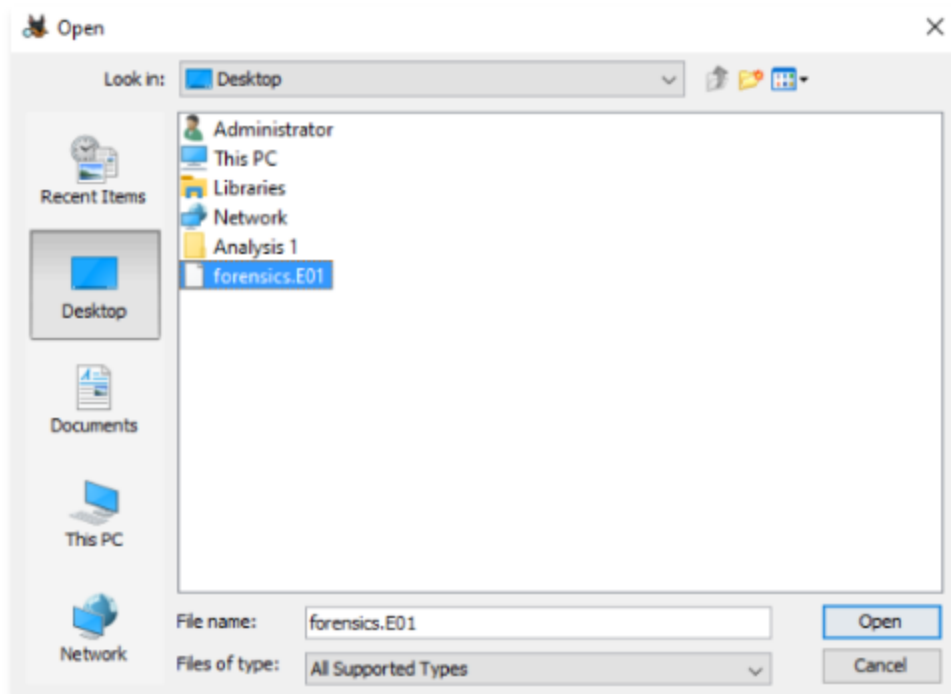
☐ Use existing host

< Back **Next >** Finish Cancel Help

7. Select **Disk Image or VM File**. Click **Next**.



8. For the data source **Path**, select **Browse**. Select **Desktop** in the left pane and open **forensics.E01**.



9. Click **Next**.

Add Data Source

Steps

1. Select Host

2. Select Data Source Type

3. **Select Data Source**

4. Configure Ingest

5. Add Data Source

Select Data Source

Path:

C:\Users\Administrator\Desktop\forensics.E01

Browse

☐ Ignore orphan files in FAT file systems

Time zone:

(GMT+0:00) UTC

Sector size:

Auto Detect

Hash Values (optional):

MD5:

SHA-1:

SHA-256:

NOTE: These values will not be validated when the data source is added.

< Back

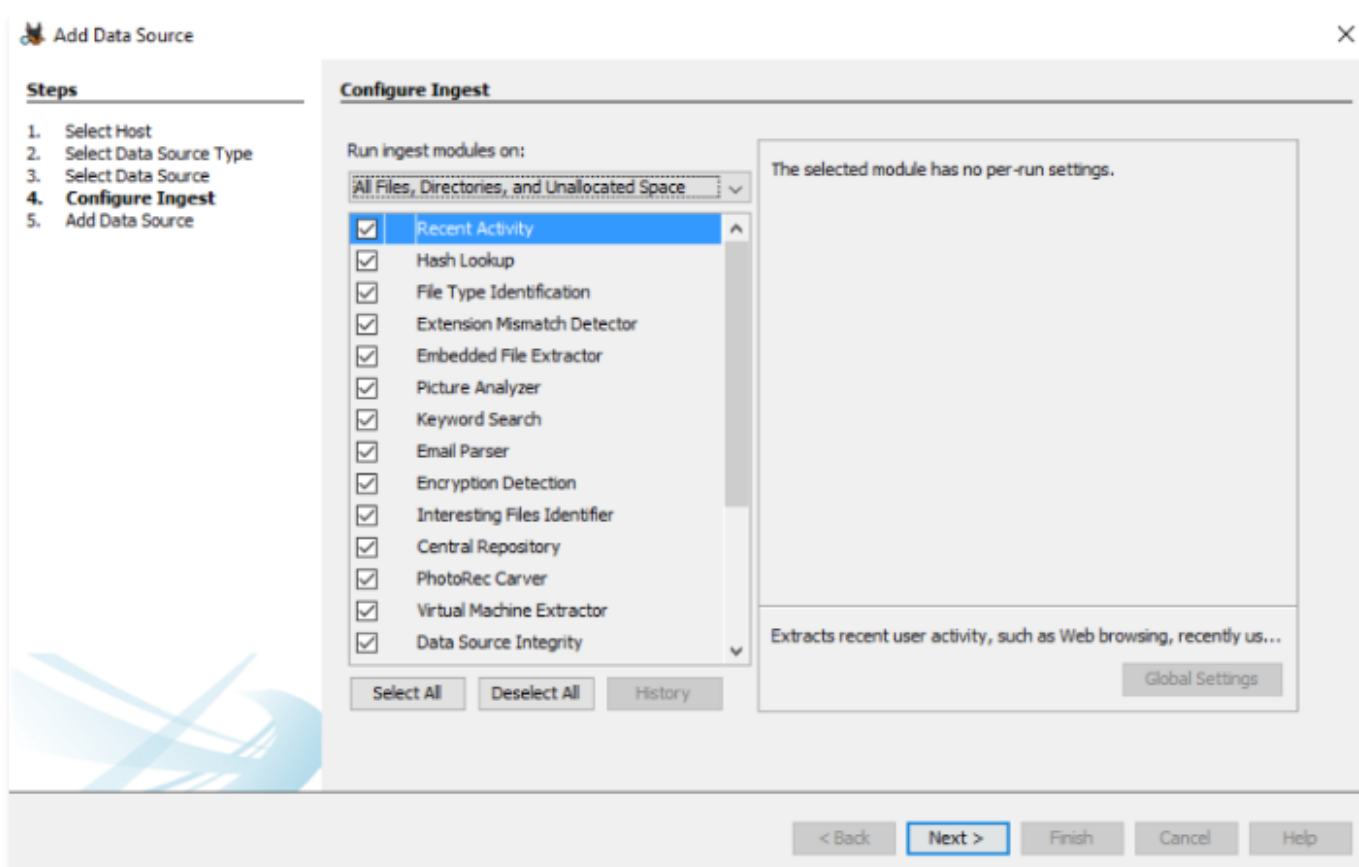
Next >

Finish

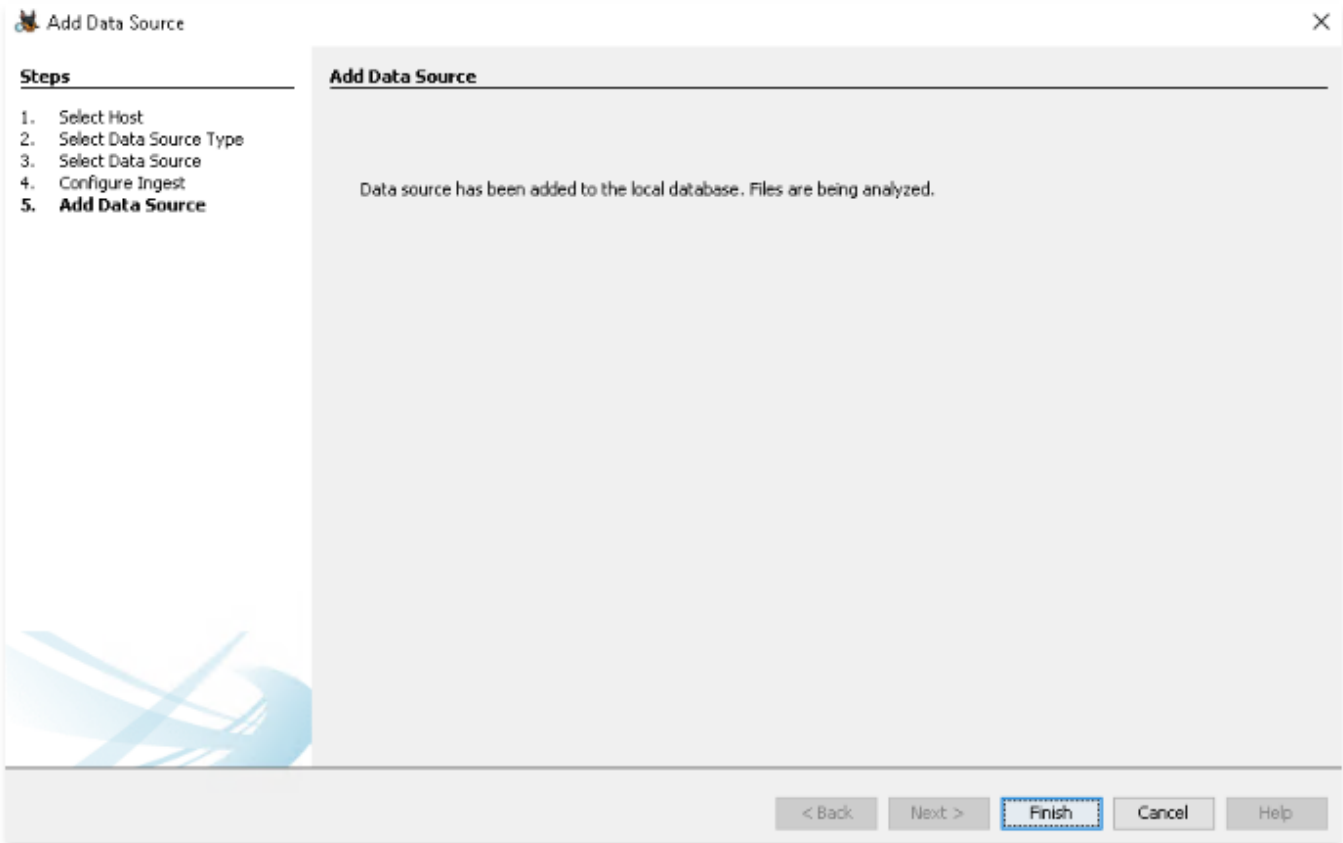
Cancel

Help

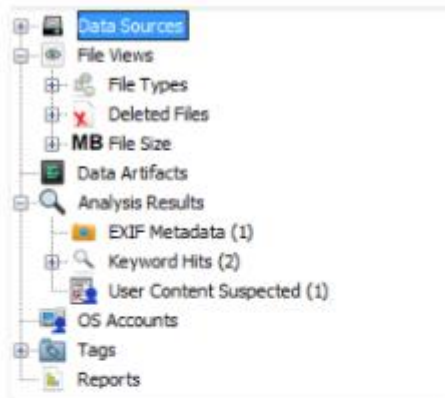
10. Click **Next**.



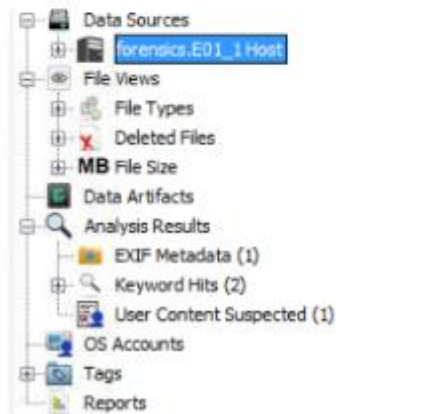
11. Click **Finish**.



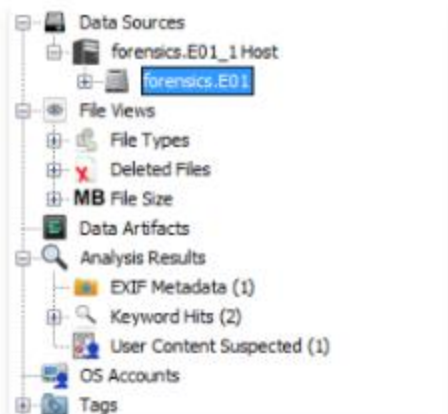
12. Expand **Data Sources** in the left pane.



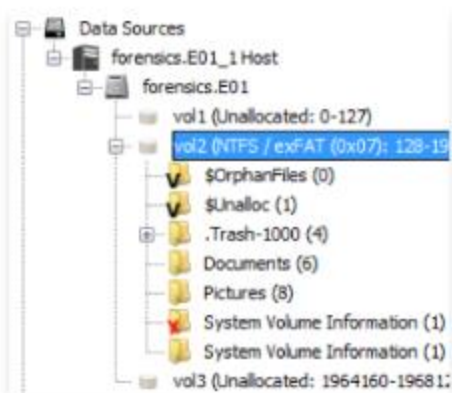
13. Click **forensics.E01_1 Host**.



14. Click **forensics.E01**.



15. Click **vol2**.



16. Select **Documents**. Right click **secret.txt** and select **Extract File(s)**. Save to desktop as **secret.txt**.

The screenshot displays a forensic analysis tool interface. On the left, a tree view shows the file system structure under 'Data Sources'. The 'Documents' folder is selected. The main pane shows a list of files, including 'secret.txt'. A right-click context menu is open over 'secret.txt', with 'Extract File(s)' highlighted. The bottom right shows a 'Text Source' dropdown set to 'File Text'.

Name	S	C	O	Modified Time	Change Time	Access Time
.secret.txt.kate-swp				2023-05-16 12:35:50 UTC	0000-00-00 00:00:00	2023-05-16 12:35:38
goodbye.txt			0	2023-05-15 14:34:58 UTC	0000-00-00 00:00:00	2023-05-16 05:12:28
hello.txt			0	2023-05-15 14:35:04 UTC	0000-00-00 00:00:00	2023-05-16 05:12:28
numbers.txt			0	2023-05-15 14:34:04 UTC	0000-00-00 00:00:00	2023-05-16 05:12:28
secret.txt				2023-05-16 12:35:51 UTC	0000-00-00 00:00:00	2023-05-16 12:35:52
work.txt				2023-05-15 14:34:42 UTC	0000-00-00 00:00:00	2023-05-16 05:12:28

Context Menu Options:

- View File in Timeline...
- View Item in New Window
- Open in External Viewer Ctrl+E
- Extract File(s)**
- Export Selected Rows to CSV
- Add File Tag
- Remove File Tag
- Add File to Hash Set
- Properties

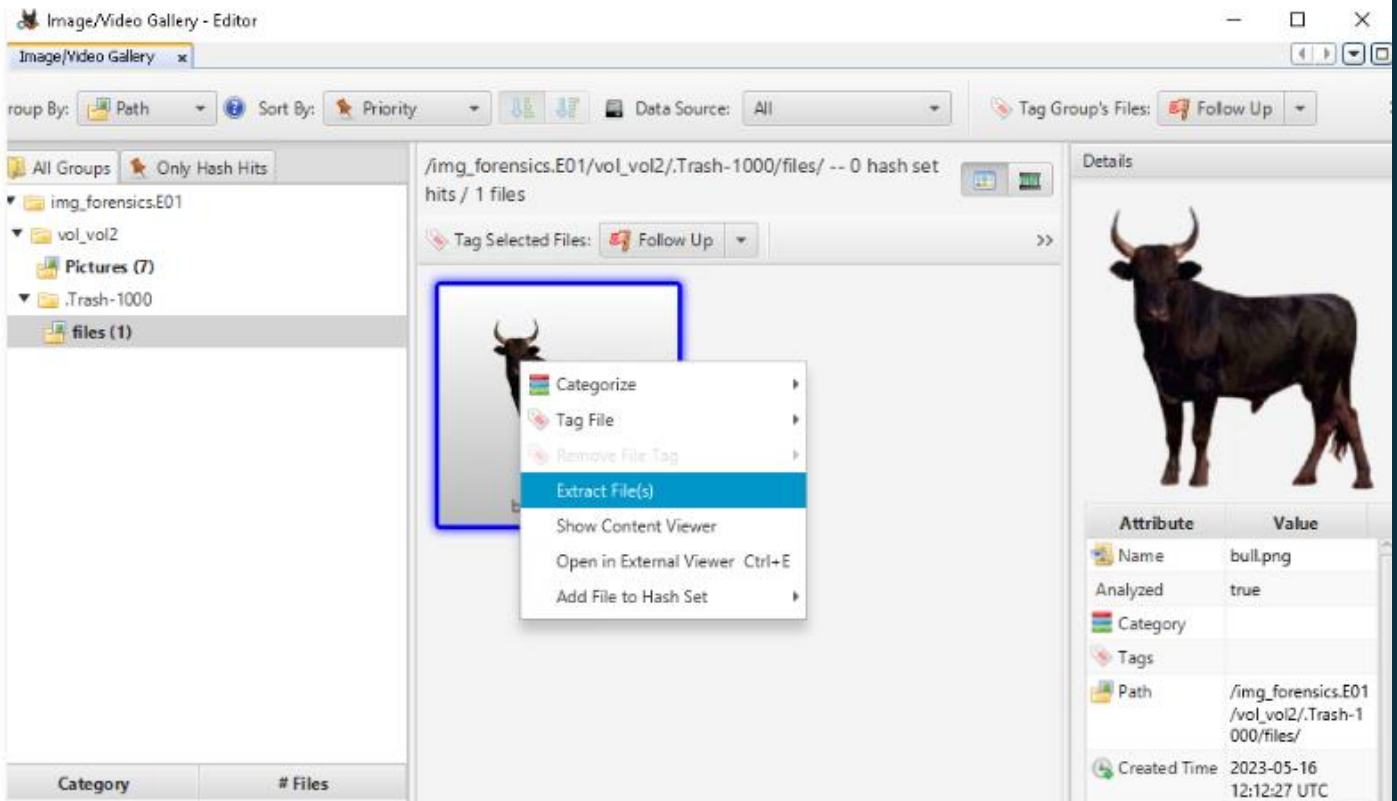
Page: 1

Text Source: File Text

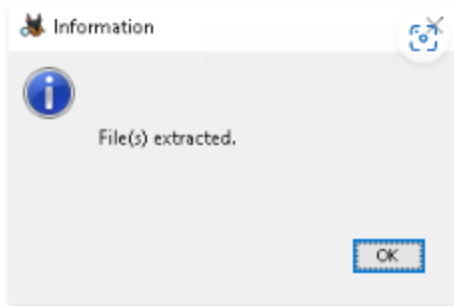
17. Click Images/Videos at the top.



18. Select **img_forensics.E01** → **vol_vol2** → **.Trash-1000** → **files**. Right click **bull.png** and **Extract File(s)**. Save to desktop as **bull.png**.



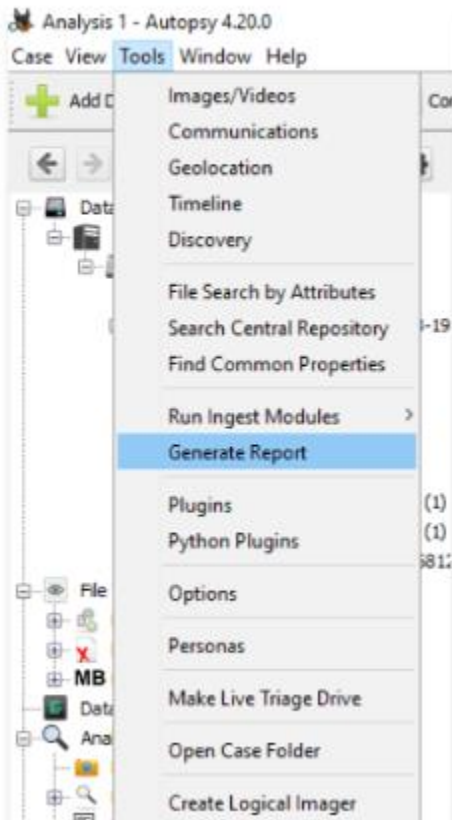
19. Click **OK** on the popup.



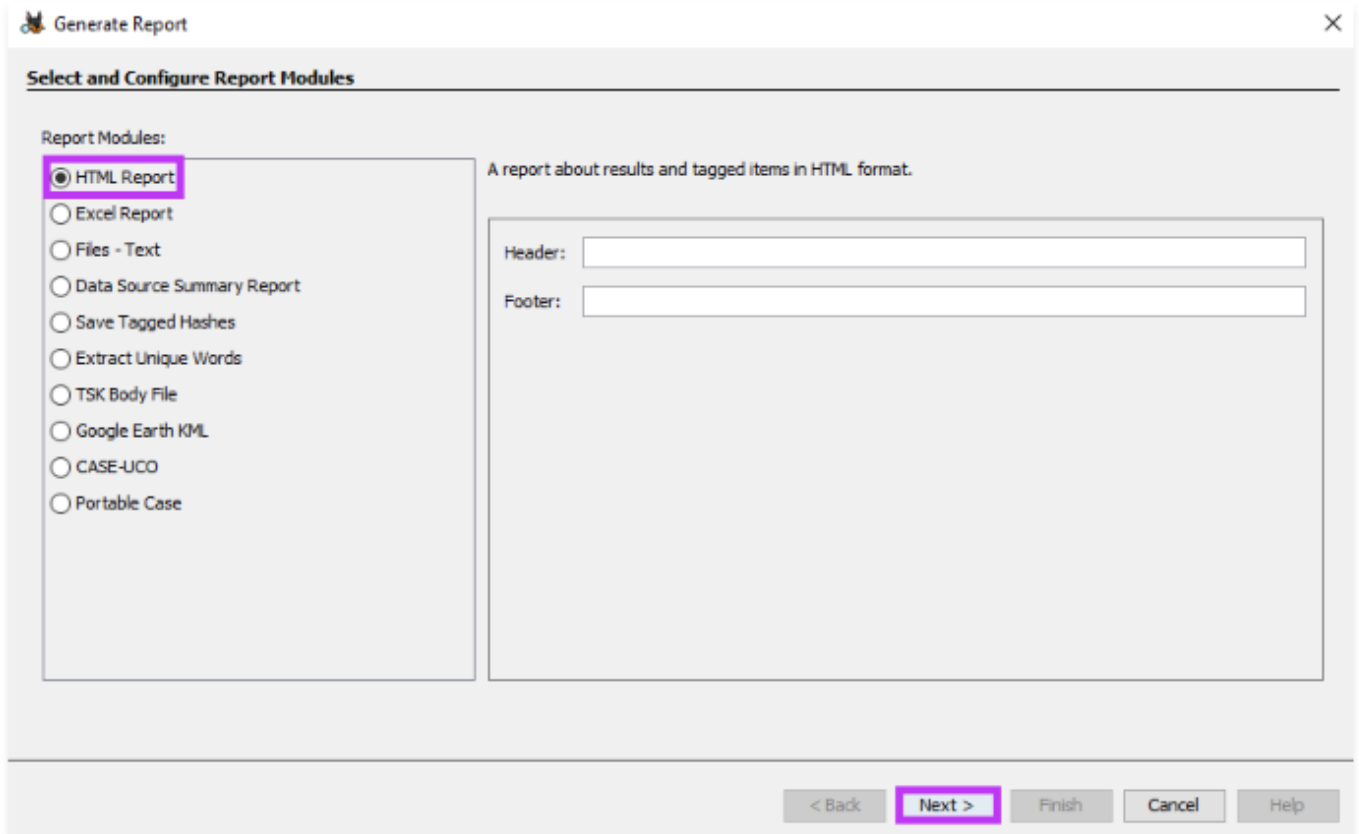
20. Close the **Image/Video Gallery - Editor**.

Task 3: Generate a report with Autopsy

1. At the top, select **Tools** and then **Generate Report**.



2. Select **HTML report**. Click **Next**.



The image shows a 'Generate Report' dialog box with a title bar containing a cat icon and a close button. The main area is titled 'Select and Configure Report Modules'. On the left, under 'Report Modules:', there is a list of radio buttons. The first option, 'HTML Report', is selected and highlighted with a red rectangle. Below it are 'Excel Report', 'Files - Text', 'Data Source Summary Report', 'Save Tagged Hashes', 'Extract Unique Words', 'TSK Body File', 'Google Earth KML', 'CASE-UCO', and 'Portable Case'. To the right of this list, there is a description: 'A report about results and tagged items in HTML format.' Below this description are two text input fields labeled 'Header:' and 'Footer:'. At the bottom of the dialog, there is a row of five buttons: '< Back', 'Next >' (highlighted with a red rectangle), 'Finish', 'Cancel', and 'Help'.

Generate Report

Select and Configure Report Modules

Report Modules:

- ☒ HTML Report
- ☐ Excel Report
- ☐ Files - Text
- ☐ Data Source Summary Report
- ☐ Save Tagged Hashes
- ☐ Extract Unique Words
- ☐ TSK Body File
- ☐ Google Earth KML
- ☐ CASE-UCO
- ☐ Portable Case

A report about results and tagged items in HTML format.

Header:

Footer:

< Back Next > Finish Cancel Help

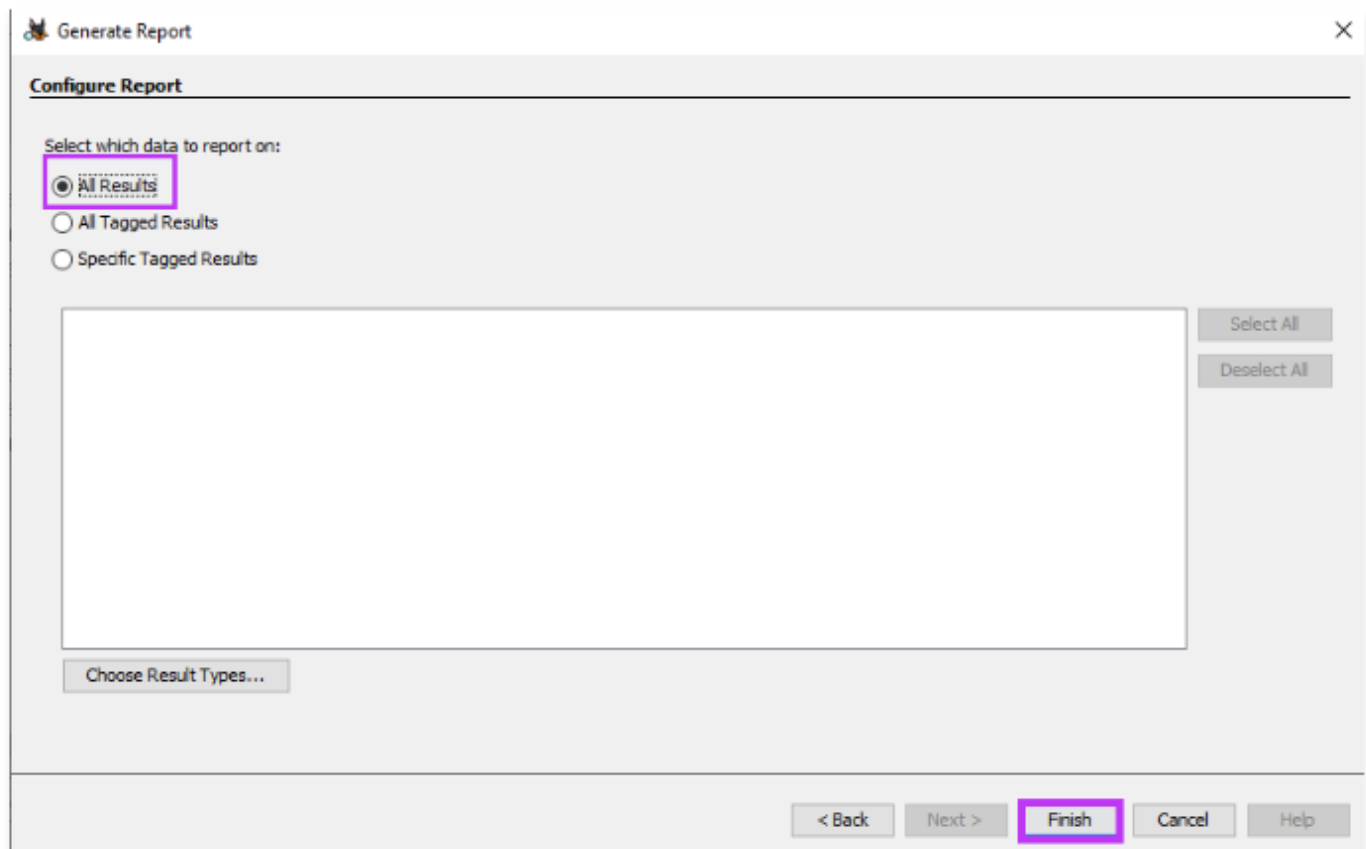
3. Check **forensics.E01**. Click **Next**.

Generate Report ×

Select which data source(s) to include

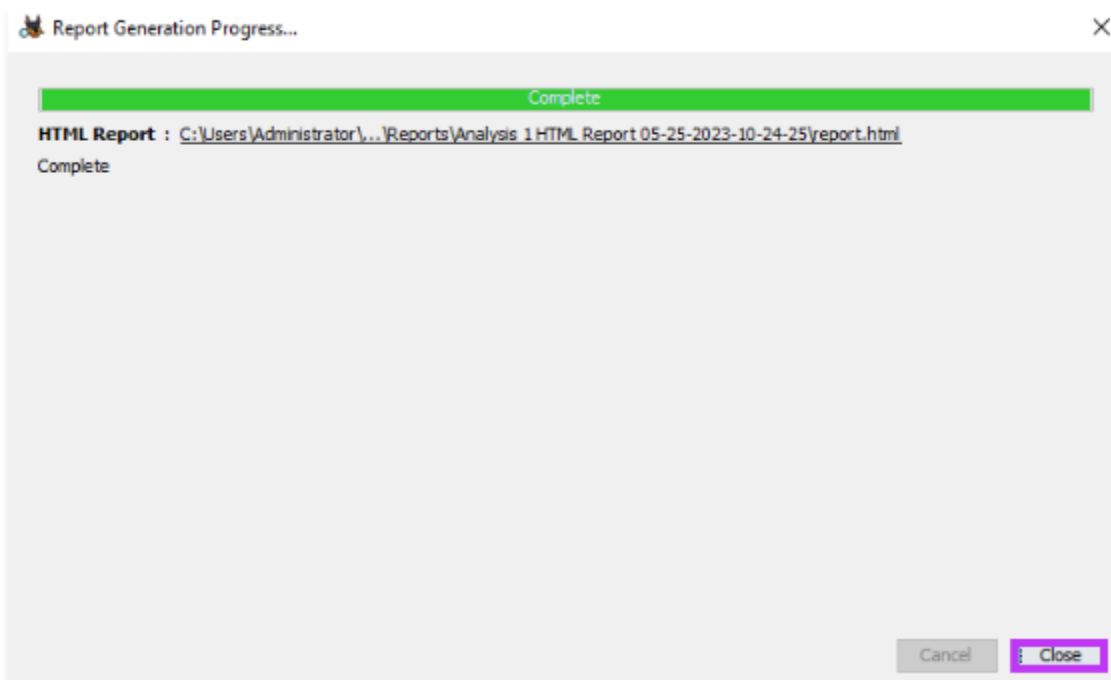
☒ forensics.E01

4. Select **All Results**. Click **Finish**.



The 'Generate Report' dialog box is shown with the 'Configure Report' tab selected. Under the heading 'Select which data to report on:', the 'All Results' radio button is selected and highlighted with a red rectangle. Below this are two unselected radio buttons: 'All Tagged Results' and 'Specific Tagged Results'. A large empty rectangular box is present below the radio buttons. To the right of this box are two buttons: 'Select All' and 'Deselect All'. At the bottom left of the dialog is a button labeled 'Choose Result Types...'. At the bottom right, there is a row of five buttons: '< Back', 'Next >', 'Finish' (highlighted with a red rectangle), 'Cancel', and 'Help'.

5. Click **Close**.



The 'Report Generation Progress...' dialog box is shown. At the top, there is a green progress bar that is completely filled, with the word 'Complete' centered above it. Below the progress bar, the text 'HTML Report : <C:\Users\Administrator\...Reports\Analysis 1 HTML Report 05-25-2023-10-24-25\report.html>' is displayed, followed by the word 'Complete' on the next line. At the bottom right, there are two buttons: 'Cancel' and 'Close' (highlighted with a red rectangle).

6. Go to the location the report is saved in. Double click the report.html link to view the autopsy report.

Report Navigation

- Case Summary
- ★ Data Source Usage (1)
- 📷 EXIF Metadata (1)
- 🔍 Keyword Hits (2)
- ★ Tagged Files (0)
- ★ Tagged Images (0)
- ★ Tagged Results (0)
- 🚨 User Content Suspected (1)

Autopsy Forensic Report

HTML Report Generated on 2023/07/06 19:20:25

Case:	Analysis 1
Case Number:	001
Number of data sources in case:	1
Examiner:	Jay Doe

Image Information:

forensics.E01

Timezone:	UTC
Path:	C:\Users\Administrator\Desktop\forensics.E01

Software Information:

Autopsy Version:	4.20.0
Android Analyzer Module:	4.20.0
Android Analyzer (aLEAPP) Module:	4.20.0

7. The lab is now complete. To end the lab, click **Delete VM** in the right pane.